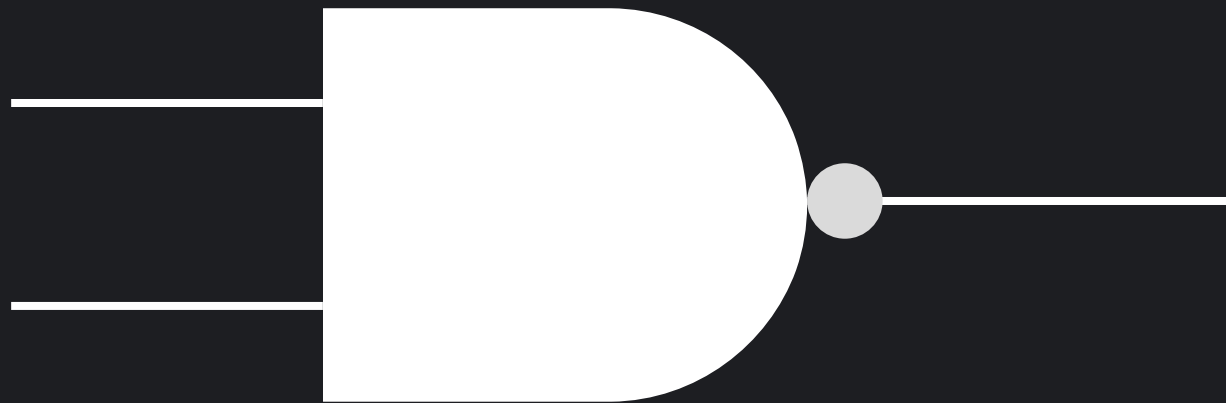


NAND GATE



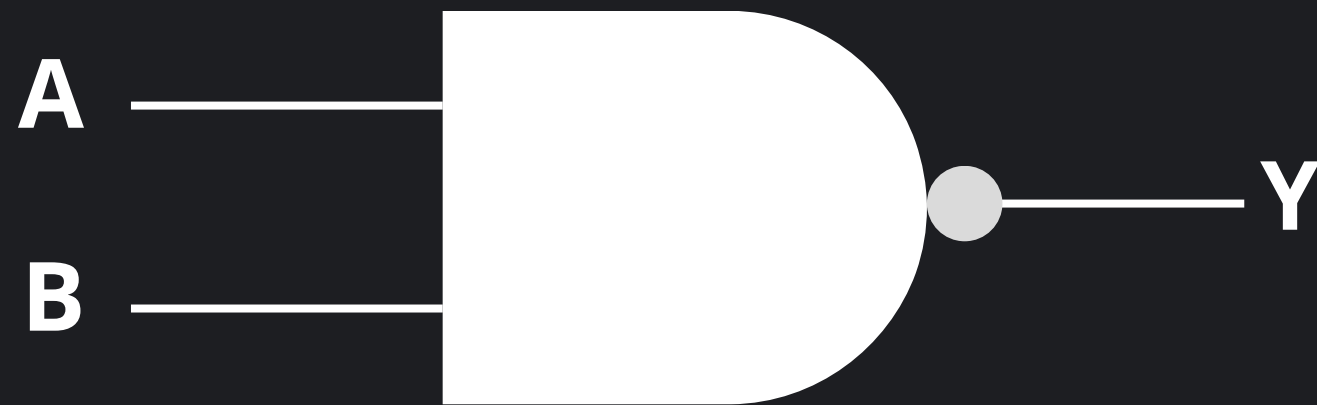
What is a NAND Gate?

A NAND gate, short for Not AND, is a basic logic gate that produces the opposite output of an AND gate. It typically has two inputs (but can have more), and its output is the negation of the AND operation.

If both inputs are 1, the output is 0.

If any input is 0 (meaning not all inputs are 1), the output is 1.

Truth Table



A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Function of the AND Gate

It performs an AND operation on the inputs and then applies a NOT operation to invert the result.

The main function of a NAND gate is to combine two or more input signals and produce an output signal according to this rule

Output = 0 only when all inputs = 1

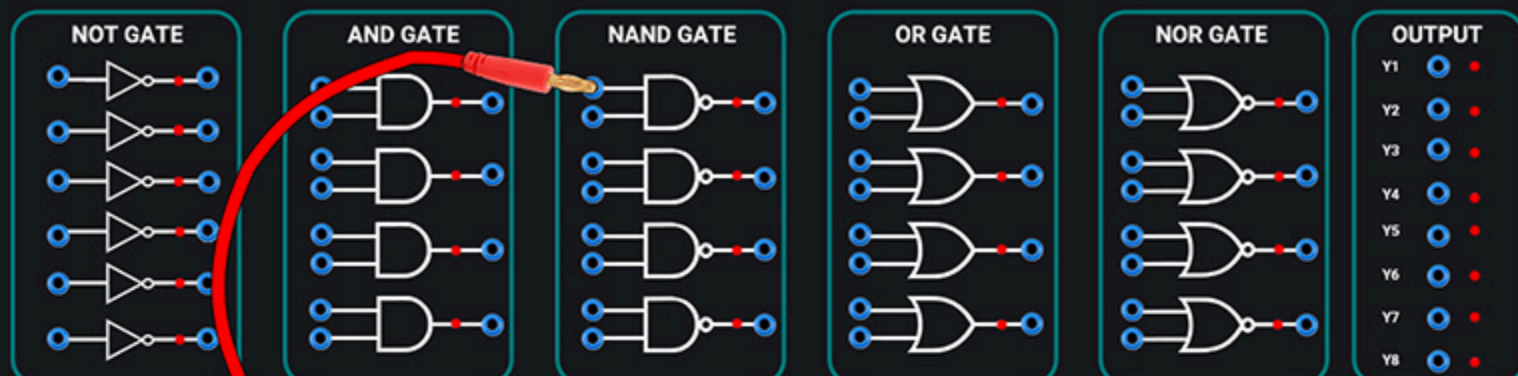
If any input = 0 $\rightarrow$ Output = 1

Summary

The NAND gate is a fundamental digital logic circuit, resulting from combining an AND gate with a NOT gate (inverter).

It performs logical NOT-AND ($\overline{A \cdot B}$), producing a LOW (0) output only when all inputs are HIGH (1).

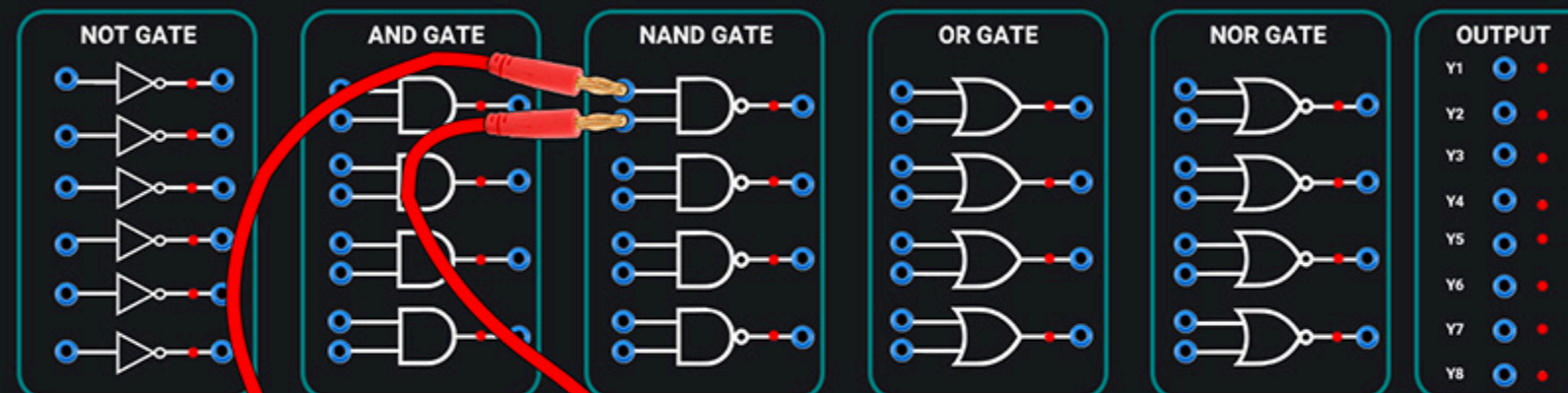
SMART DIGITAL CIRCUITS BOARD



1

Connect a wire from the first input (top blue terminal) to switch A in the INPUT section.

SMART DIGITAL CIRCUITS BOARD



2

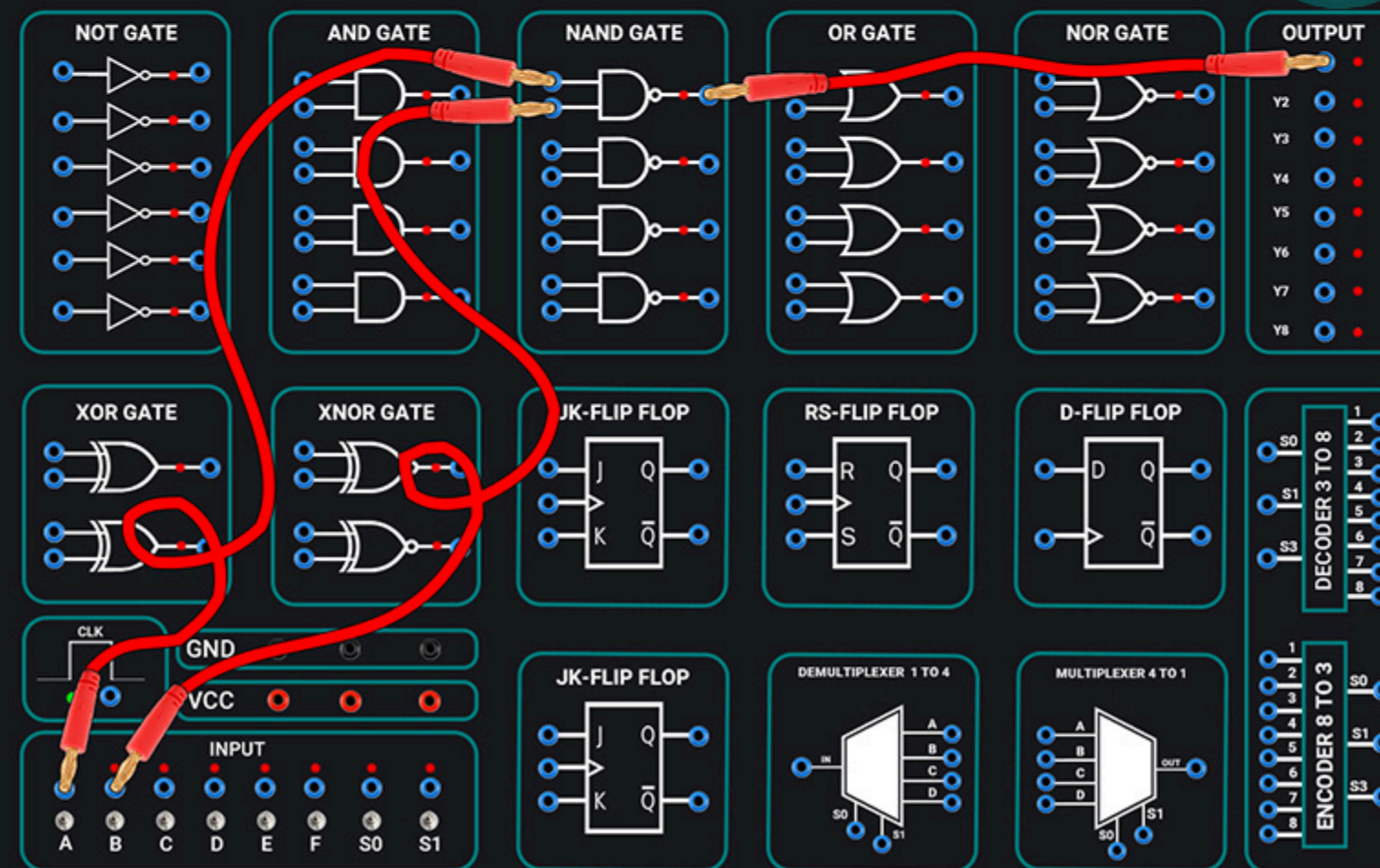
Connect a wire from the second input (bottom blue terminal) to switch B in the INPUT section

Connect the NAND Gate

Connect a wire from the gate output to terminal Y1 (or any point from Y1 to Y8) in the OUTPUT section at the top right.

3

SMART DIGITAL CIRCUITS BOARD



SMART DIGITAL CIRCUITS BOARD

